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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM05.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proofs of (1.1)–(1.4), page 93: the proofs are added.
2. **Misprint.** Statement of (1.5), page 93: the second "planar" is removed.
3. **Degenerate case.** Statement of (4.3), page 98: the hypothesis $d+n > 0$ is added.
4. **Degenerate case.** Statement of (4.4), page 98: the hypothesis $p > 0$ is added.
5. **Misprint.** Statement of (4.5), page 99: θ_2 becomes θ_1.
6. **Degenerate case.** Statement of (5.1), page 100: the hypothesis $p \geq 2$ is added.
7. **Statement mistake.** Statement of (8.11), page 113: r must not be a whole component of Σ homeomorphic to a sphere.

8. **Missing proof.** Proof of (8.11), page 113: a proof is added.
9. **Statement mistake.** Statement of (8.12), page 113: the drawing must meet every component of Σ homeomorphic to a sphere.
10. **Missing proof.** Proof of (8.14), page 113: the choice of the drawings and the details of the last steps are added.

[p. 92] Graph Minors. V. Excluding a Planar Graph

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We prove that for every planar graph H there is a number w such that every graph with no minor isomorphic to H can be constructed from graphs with at most w vertices, by piecing them together in a tree structure. This has several consequences; for example, it implies that: (i) if $\mathscr{A}$ is a set of graphs such that no member is isomorphic to a minor of another, and some member of $\mathscr{A}$ is planar, then $\mathscr{A}$ is finite; (ii) for every fixed planar graph H there is a polynomial time algorithm to test if an arbitrary graph has a minor isomorphic to H; (iii) there is a generalization of a theorem of Erdös and Pósa (concerning the maximum number of disjoint circuits in a graph) to planar structures other than circuits.

1. Introduction

Graphs in this paper are finite, and may have loops and multiple edges. A graph is a minor of another if the first can be obtained by contraction from a subgraph of the second.

We begin with a definition of the "tree-width" of a graph. This concept has been discussed in other papers of this series, but for the reader's convenience we define it again here. A *tree-decomposition* (T, χ) of a graph G is a tree T together with a family $\chi = (X_t : t \in V(T))$ of subsets of $V(G)$, such that

(i) $\bigcup(X_t : t \in V(T)) = V(G)$;

(ii) for each edge e of G there exists $t \in V(T)$ such that e has both ends (or its end, in the case of a loop) in X_t;

[p. 93] (iii) for $t, t', t'' \in V(T)$, if t' lies on the path of T between t and t'', then $X_t \cap X_{t''} \subseteq X_{t'}$.

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The width of the tree-decomposition is $\max(|X_t|-1 : t \in V(T))$; and the *tree-width* of G is the minimum $w \geq 0$ such that G has a tree-decomposition of width $\leq w$.

The following results are easy to prove.

(1.1) *The simple graphs of tree-width ≤ 1 are precisely the forests.*

(1.2) *The graphs of tree-width ≤ 2 are precisely the series-parallel graphs.*

(1.3) *If H is a minor of G then the tree-width of H is no greater than the tree-width of G.*

(1.4) *For $n \geq 1$, the complete graph K_n has tree-width $n-1$, and for $\theta \geq 2$ the θ-grid has tree-width θ.*

(When $\theta \geq 2$ is an integer, the θ-grid is the simple graph with vertex set $\{v_{ij} : 1 \leq i, j \leq \theta\}$, in which v_{ij} and $v_{i'j'}$ are adjacent if $|i-i'|+|j-j'| = 1$.)

We give the proofs of (1.1)–(1.4). The *simple graph* of a graph is obtained by deleting its loops and replacing each class of parallel edges by one edge. A graph and its simple graph have the same tree-decompositions, since (ii) holds for a loop at v as soon as v is in some X_t, which is true by (i). Let (T, χ) be a tree-decomposition of a graph G, and let $t \in V(T)$ have valency 1 in T, with neighbour s. We shall use two facts. First, if $X_t \subseteq X_s$, then deleting t from T and X_t from χ gives a tree-decomposition of G, because X_s can play the part of X_t in (i) and (ii), and the paths of $T \setminus t$ are paths of T. Second, if $v \in X_t - X_s$, then every neighbour x of v is in X_t. For by (ii) some X_q contains v and x, and if $q \neq t$ then s lies on the path of T between t and q, and so $v \in X_t \cap X_q \subseteq X_s$ by (iii), which is false.

Proof of (1.1). Suppose first that G has tree-width ≤ 1 and has a circuit C. Replacing each X_t by $X_t \cap V(C)$ gives a tree-decomposition of C of width ≤ 1, since (i)–(iii) for C follow from (i)–(iii) for G. In C every vertex has two distinct neighbours, since G is simple. We show by induction on $|V(T)|$ that no non-null graph G' in which every vertex has two distinct neighbours has a tree-decomposition (T, χ) of width ≤ 1. If $|V(T)| = 1$, then by (i) the only bag is $V(G')$, which contains a vertex of G' and its two neighbours. This is impossible. Otherwise let t have valency 1 in T, with neighbour s. If some $v \in X_t - X_s$, then X_t contains v and its two neighbours, by the second fact, which is impossible. So $X_t \subseteq X_s$, and the first fact gives a tree-decomposition of G' of width ≤ 1 with fewer bags, contrary to induction. Hence G is a forest. Conversely, we show by induction on $|V(G)|$ that every forest G has tree-width ≤ 1. If $V(G) = \emptyset$, a single empty bag will do. Otherwise some vertex v of G has at most one neighbour (for instance an end of a longest path of G), and by induction $G \setminus v$ has a tree-decomposition

(T, χ) of width ≤ 1. Choose $t_0 \in V(T)$ such that X_{t_0} contains the neighbour of v, if there is one, by (i). Add to T a new vertex t' adjacent to t_0, with $X_{t'}$ consisting of v and its neighbours. This is a tree-decomposition of G of width ≤ 1. For (iii), v is in no other bag, and $X_{t'} - \{v\} \subseteq X_{t_0}$, where t_0 lies on the path of T from t' to every other vertex.

Proof of (1.2). The paper does not define series-parallel graphs. We use the usual definition by series and parallel reductions, in the following form. A graph G is *series-parallel* if its simple graph can be reduced to the null graph by repeatedly deleting a vertex with at most two neighbours and, if it has two, joining them by an edge if they are not already adjacent. We call this a *reduction* at the vertex.

We first show that some bag X_t contains a vertex of G and all its neighbours, if G is non-null. We use induction on $|V(T)|$. If $|V(T)| = 1$, any vertex will do, since the only bag is $V(G)$ by (i). Otherwise let t have valency 1 in T, with neighbour s. If $X_t \subseteq X_s$ we apply induction to the tree-decomposition given by the first fact. Otherwise we take any $v \in X_t - X_s$, by the second fact.

Now suppose that G has a tree-decomposition of width ≤ 2. We show by induction on $|V(G)|$ that G is series-parallel. By the remark above we may assume that G is simple, and that it is non-null. Some bag X_t contains a vertex v and all its neighbours, and so v has at most two neighbours, since $|X_t| \leq 3$. Let G' be the graph obtained by the reduction at v. Removing v from every bag gives a tree-decomposition of G' of width ≤ 2. Indeed (i) and (iii) hold. For (ii), the edges of G' that are edges of G are not incident with v, and the new edge of G', if any, has both ends in $X_t - \{v\}$. By induction G' is series-parallel, and therefore so is G. Conversely, suppose that G is series-parallel. Again we may assume that G is simple. We use induction on the number of reductions. Let the first reduction be at v, giving G'. By induction G' has a tree-decomposition (T, χ) of width ≤ 2, and it is also a tree-decomposition of $G \setminus v$, since every edge of $G \setminus v$ is an edge of G'. Some bag X_{t_0} contains all neighbours of v. If v has at most one neighbour this follows from (i). If v has two neighbours, they are adjacent in G', and it follows from (ii). Add to T a new vertex t' adjacent to t_0, with $X_{t'}$ consisting of v and its neighbours. As in the proof of (1.1), this is a tree-decomposition of G, and its width is ≤ 2.

Proof of (1.3). Let (T, χ) be a tree-decomposition of G whose width w is the tree-width of G. We show that H has a tree-decomposition of width $\leq w$. Since H is a minor of G, there are disjoint non-empty sets $Y(v) \subseteq V(G)$ $(v \in V(H))$, each inducing a connected subgraph of G, such that for every edge of H with distinct ends u, v, some edge of G has one end

in $Y(u)$ and the other in $Y(v)$. (Take for $Y(v)$ the set of vertices of G that are contracted into v.) For $t \in V(T)$ let X'_t be the set of all $v \in V(H)$ with $Y(v) \cap X_t \neq \emptyset$. Choosing for each $v \in X'_t$ a vertex in $Y(v) \cap X_t$ gives distinct vertices for distinct v, since the sets $Y(v)$ are disjoint, and so $|X'_t| \leq |X_t|$. We claim that $(T, (X'_t))$ is a tree-decomposition of H. (i) holds since each $Y(v)$ is non-empty. For (ii), by the remark above we need only consider edges with two ends. If u, v are the ends of an edge of H, some edge of G has ends $x \in Y(u)$ and $y \in Y(v)$, and some X_t contains x and y, and then $u, v \in X'_t$. For (iii), we first show:

(1) *Let K be a connected subgraph of G, and let $a, b, c \in V(T)$, where c lies on the path of T between a and b. If K has a vertex in X_a and a vertex in X_b, then K has a vertex in X_c.*

Let $x \in V(K) \cap X_a$. We show, by induction on the length of a walk of K from x to y, that for every $b \in V(T)$ with $y \in X_b$, and every c on the path of T between a and b, some vertex of K is in X_c. If the walk has length 0 then $y = x$, and $x \in X_a \cap X_b \subseteq X_c$ by (iii). Otherwise let y' be the vertex of the walk before y. By (ii) some X_q contains y' and y. The path of T between a and b is contained in the union of the paths between a and q and between q and b, and so c lies on one of them. If c lies on the path between a and q, we apply induction to the walk from x to y', since $y' \in X_q$. If c lies on the path between q and b, then $y \in X_q \cap X_b \subseteq X_c$ by (iii). This proves (1).

Now if $v \in X'_a \cap X'_b$ and c lies on the path of T between a and b, then by (1) applied to the subgraph induced on $Y(v)$, some vertex of $Y(v)$ is in X_c, and so $v \in X'_c$. This proves (iii), and (1.3) follows.

Proof of (1.4). A single bag $V(K_n)$ shows that K_n has tree-width $\leq n-1$. Conversely, in a tree-decomposition of K_n, some bag contains a vertex and all its neighbours, as shown in the proof of (1.2), and so it contains all n vertices. Hence K_n has tree-width $n-1$.

Now let $\theta \geq 2$, and for $1 \leq i, j \leq \theta$ let $r(v_{ij}) = \theta(i-1) + j - 1$, so that r maps the vertices of the θ-grid one-to-one onto $\{0, \ldots, \theta^2 - 1\}$, and adjacent vertices have values of r differing by 1 or θ. Let $L = \theta^2 - \theta$, let T be the path with vertices $0, 1, \ldots, L-1$ in order, and for $0 \leq k \leq L-1$ let X_k be the set of vertices v with $k \leq r(v) \leq k + \theta$. Each X_k has at most $\theta + 1$ vertices. A vertex v is in X_k for $k = \min(r(v), L-1)$, and the ends x, y of an edge, with $r(x) < r(y)$, are both in X_k for $k = \min(r(x), L-1)$, since $r(y) \leq \min(r(x) + \theta, \theta^2 - 1)$. The bags containing v are those with $r(v) - \theta \leq k \leq r(v)$, and they form a subpath of T. So this is a tree-decomposition of width $\leq \theta$.

For the converse we need two more facts.

(2) *Let A_j $(j \in J)$ be non-empty sets of vertices of a tree T, each*

containing the path of T between any two of its members, such that $A_i \cap A_j \neq \emptyset$ for all $i, j \in J$. Then some vertex of T is in every A_j.

We use induction on $|V(T)|$. If $|V(T)| = 1$ this holds since each A_j is non-empty. Otherwise let t have valency 1 in T, with neighbour s. If $A_i \subseteq \{t\}$ for some i, then $t \in A_j$ for every j, since $A_i \cap A_j \neq \emptyset$. Otherwise each A_j has a vertex other than t, and so if $t \in A_j$ then $s \in A_j$, since s lies on the path from t to that vertex. Hence the sets $A_j - \{t\}$ are non-empty, each contains the path of $T \setminus t$ between any two of its members, and any two of them meet, since if $t \in A_i \cap A_j$ then $s \in A_i \cap A_j$. By induction some vertex of $T \setminus t$ is in all of them. This proves (2).

(3) *Let (T, χ) be a tree-decomposition of G, and let M_j $(j \in J)$ be connected subgraphs of G such that for all $i, j \in J$, either M_i and M_j have a common vertex or some edge of G joins them. Then some bag X_t contains a vertex of every M_j.*

For $j \in J$, let A_j be the set of $t \in V(T)$ such that X_t contains a vertex of M_j. It is non-empty by (i), and it contains the path of T between any two of its members, by (1). For $i, j \in J$, a bag containing a common vertex of M_i and M_j (which exists by (i)), or both ends of an edge joining them (which exists by (ii)), is in $A_i \cap A_j$. So (3) follows from (2).

Now let (T, χ) be a tree-decomposition of the θ-grid. Let R be the subgraph induced on $\{v_{1j} : 1 \leq j \leq \theta\}$, and let C be the subgraph induced on $\{v_{i1} : 2 \leq i \leq \theta\}$. For $2 \leq i, j \leq \theta$, let Q_{ij} be the subgraph induced on the union of $\{v_{ij'} : 2 \leq j' \leq \theta\}$ and $\{v_{i'j} : 2 \leq i' \leq \theta\}$. These subgraphs are connected, and any two of them have a common vertex or are joined by an edge. For $v_{11} \in V(R)$ is adjacent to $v_{21} \in V(C)$. For $2 \leq i, j \leq \theta$, the vertex v_{1j} of R is adjacent to $v_{2j} \in V(Q_{ij})$, and the vertex v_{i1} of C is adjacent to $v_{i2} \in V(Q_{ij})$. Finally, for $2 \leq i, j, i', j' \leq \theta$, the vertex $v_{ij'}$ belongs to both Q_{ij} and $Q_{i'j'}$. By (3), some bag X_t contains a vertex x of R, a vertex y of C, and a vertex of every Q_{ij}. Let S be the set of vertices $v_{ij} \in X_t$ with $i, j \geq 2$. If for every $i \geq 2$ some v_{ij} with $j \geq 2$ is in X_t, then $|S| \geq \theta - 1$. Otherwise there exists $i \geq 2$ such that no v_{ij} with $j \geq 2$ is in X_t. Then for each $j \geq 2$, the vertex of Q_{ij} in X_t is $v_{i'j}$ for some $i' \geq 2$, and so again $|S| \geq \theta - 1$. The vertices x and y are distinct and not in S. Hence $|X_t| \geq \theta + 1$, and the θ-grid has tree-width θ.[F1]

The principal result of this paper is

(1.5) *For every planar graph H, there is a number w such that every*

[F1] **Missing proof.** The paper says that (1.1)–(1.4) are easy to prove and gives no proofs. The arguments above are the ones used in the formalization.

~~*planar*~~ *graph with no minor isomorphic to H has tree-width $\leq w$.*[F2]

We remark that if H is non-planar there is no such number w: because, for example, the θ-grids have arbitrarily large tree-width and yet have no minor isomorphic to H.

The main step in the proof of (1.5) is the following, which seems to be of some independent interest.

(1.6) *For any positive integer θ there exists an integer θ' such that if G has subgraphs H and K, each with $\geq \theta'$ components, and if each component of H meets each component of K, then G contains a θ-grid minor.*

In earlier papers of this series, (1.5) was used to show the following.

(1.7) [11]. *If $G_1, G_2, \ldots$ is a countable sequence of graphs and G_1 is planar, then there exist i, j with $j > i$ such that G_i is isomorphic to a minor of G_j.*

(This is a partial result towards an unpublished conjecture of Wagner, that (1.7) is true even if G_1 is not planar.)

[p. 94] (1.8) [10]. *If H is any fixed planar graph, there is a polynomial time algorithm to test if an arbitrary graph has a minor isomorphic to H.*

(In a later paper we shall show that this is true even for non-planar H.) In Section 8 we give a new application of (1.5), generalizing the Erdös-Pósa theorem on the maximum number of disjoint circuits in a graph.

Let us clarify some terminology. It will sometimes be convenient to identify a subset of the vertex set of a graph with the corresponding subgraph with no edges, but this should cause no confusion. Two subgraphs of a graph are *disjoint* if they have no common vertices, and otherwise they *meet*. Subgraphs $G_1, \ldots, G_n$ of a graph are disjoint if every two of them are disjoint. A *path* has at least one vertex, and has no "repeated" vertices. Every path has an *initial vertex* and a *terminal vertex*. If A, B are subgraphs of a graph, a *path from A to B* has its initial vertex in $V(A)$ and its terminal vertex in $V(B)$. If A, B, C are subgraphs of a graph, C *separates A and B* if every path from A to B meets C. A *separation* of a graph G is a pair (V_1, V_2), where $V_1, V_2 \subseteq V(G)$, $V_1 \cup V_2 = V(G)$, and no vertex of $V_1 - V_2$ is adjacent to any vertex of $V_2 - V_1$. If $X \subseteq V(G)$, $G \setminus X$ denotes the result of deleting X from G, and $G|X$ denotes $G \setminus (V(G) - X)$. We often abbreviate $G \setminus \{v\}$ by $G \setminus v$. A graph G is a *subdivision* of a graph H if H can be obtained from G by (repeatedly) contracting a non-loop edge incident with a vertex of valency 2.

[F2] **Misprint.** The second "planar" is a misprint. The paper proves and uses the statement for every graph: (2.1), its restatement on page 95, says "every graph", and the proof of (8.8) on page 111 applies (1.5) to all graphs.

2. The Main Result

Let us restate (1.5) to show the dependence of w upon H. For any planar graph H there is an even value of $\theta \geq 6$ such that H is isomorphic to a minor of the θ-grid. (This is easy to see, but for a proof see [1, 12].) Denote by $\theta(H)$ the smallest such value of θ.

Now let $\theta \geq 6$ be some even number. For $k \geq 2$, $n \geq 0$, define $\alpha(k, n)$ inductively by

$$\alpha(2, n) = n + 1$$
$$\alpha(k, n) = 2^{n\theta^4} + \alpha(k-1, 2^{n\theta^4} + n + 1), \quad \text{for } k \geq 3.$$

Define $\theta_1 = 2\alpha(\theta^2/2, \theta^2/2)$. Define $\phi_{\theta_1} = \theta^2/2$, and for $0 \leq k < \theta_1$ define ϕ_k inductively by

$$\phi_k = \phi_{k+1} 2^{\phi_{k+1}\theta^2}.$$

Define [p. 95]

$$\theta_2 = \phi_0 + 2\phi_1 + 2\phi_2 + \cdots + 2\phi_{\theta_1 - 1} + \phi_{\theta_1}$$
$$\theta_3 = (\theta^2/2)^{\theta_2 - 1}$$
$$\theta_4 = \theta_2 \binom{\theta_3}{\theta_2} + \frac{1}{2}\theta^2 \binom{\theta_3}{\theta^2/2}$$
$$\theta_5 = (\theta^2/2)^{\theta_4 - 1}$$
$$\theta_6 = \theta_3 \binom{\theta_5}{\theta_4} + \frac{1}{2}\theta^2 \binom{\theta_5}{\theta^2/2}$$
$$\theta_7 = \alpha(\theta_5, \theta_6)$$
$$\theta_8 = 3\theta_5(3^{\theta_5} - 1)/4$$
$$\theta_9 = \theta_7(\theta_8 + 1) + 1.$$

To emphasize the dependence of these parameters on θ, we occasionally write $\theta_9(\theta)$ instead of θ_9, etc.

(2.1) *If H is a planar graph, then every graph with no minor isomorphic to H has tree-width at most $\theta_9(\theta(H))$.*

To prove this, we observe that if G has no minor isomorphic to H then G certainly has no minor isomorphic to the θ-grid, where $\theta = \theta(H)$. It suffices therefore to prove that for $\theta \geq 6$ and even, if G has no minor isomorphic to the θ-grid then G has tree-width $\leq \theta_9$. This we shall show in (7.3). Throughout the rest of the paper, θ is a fixed even integer with $\theta \geq 6$. Let $\mathscr{F}_\theta$ be the class of all graphs with no minor isomorphic to the θ-grid.

3. Grids

The following was proved in [9, Theorem (2.9)].

(3.1) *Let $B_1, \dots, B_n$ be disjoint connected subgraphs of $G \in \mathscr{F}_\theta$, and let $V' \subseteq V(G)$. Let $r \geq 0$ be an integer. Then at least one of the following is true:*

(i) *there exists $J \subseteq \{1, \dots, n\}$ with $|J| = r$ such that for each $j \in J$ there is a path P_j in G from B_j to V', and the paths $P_j (j \in J)$ are disjoint, and each has no internal vertex in $V' \cup \bigcup_{j \in J} V(B_j)$;*

(ii) *there exists $X \subseteq V(G)$ and $J \subseteq \{1, \dots, n\}$ with $|X| + |J| \leq 2^{r\theta^4}$ such that X separates V' from every B_j $(j \in \{1, \dots, n\} - J)$.*

The object of the present section is to use (3.1) to prove the following. Note that $\alpha(k, n)$ is defined in Section 2.

[p. 96] (3.2) *Let $G \in \mathcal{F}_\theta$, and let $V_1, \dots, V_k \subseteq V(G)$, where $k \geq 2$. Let $n \geq 0$ be an integer. Then at least one of the following is true:*

(i) *there are disjoint connected subgraphs $B_1, \dots, B_n$ of G such that each B_j meets each V_i $(1 \leq i \leq k, 1 \leq j \leq n)$*

(ii) *there exists $X \subseteq V(G)$ with $|X| < \alpha(k, n)$ such that every connected subgraph of G which meets all of $V_1, \dots, V_k$ also meets X.*

Proof. We proceed by induction on k. If $k = 2$ the result follows from Menger's theorem, and we assume then that $k \geq 3$ and that the result holds for $k - 1$.

Let $C_1, \dots, C_N$ be disjoint connected subgraphs of G, each meeting all of $V_1, \dots, V_{k-1}$, chosen so that N is maximum. We may assume that $N \geq n$; for otherwise the result follows from our inductive hypothesis and the fact that $\alpha(k-1, n) \leq \alpha(k, n)$. We may assume that fewer than n of $C_1, \dots, C_N$ meet V_k since otherwise (i) holds, and so we may assume that $C_1, \dots, C_{N-n}$ do not meet V_k. We apply (3.1), and deduce that one of the following cases holds.

Case 1. There exists $J \subseteq \{1, \dots, N-n\}$ with $|J| = n$ such that for each $j \in J$ there is a path P_j from C_j to V_k, and the paths P_j $(j \in J)$ are disjoint, and each has no internal vertex in $V_k \cup \bigcup_{j \in J} V(C_j)$.

In this case (i) is true, using the subgraphs $C_j \cup P_j$ $(j \in J)$.

Case 2. There exists $X' \subseteq V(G)$ and $J \subseteq \{1, \dots, N-n\}$ with $|X'| + |J| \leq 2^{n\theta^4}$ such that X' separates V_k from every C_j $(j \in \{1, \dots, N-n\} - J)$.

In this case, let (W_1, W_2) be a separation of G with $W_1 \cap W_2 = X'$, $V_k \subseteq W_1$, and $V(C_j) \subseteq W_2$ $(j \in \{1, \dots, N-n\} - J)$. Let $W = W_1 - W_2$, let $G' = G|W$, and let $V_i' = V_i \cap W$ $(1 \leq i \leq k-1)$.

Now $V(C_j) \cap W = \emptyset$ for each $j \in \{1, \ldots, N-n\} - J$, since $V(C_j) \subseteq W_2$. By the maximality of N, the maximum number of disjoint connected subgraphs of G' each meeting all of $V'_1, \ldots, V'_{k-1}$ is at most

$$|J| + n < 2^{n\theta^4} + n + 1$$

and so by our inductive hypothesis there exists $X'' \subseteq V(G')$ with

$$|X''| < \alpha(k-1, 2^{n\theta^4} + n + 1)$$

such that every connected subgraph of G' which meets all of $V'_1, \ldots, V'_{k-1}$ also meets X''. But then every connected subgraph B of G which meets all of $V_1, \ldots, V_k$ also meets $X' \cup X''$. For if such a B does not meet X', then $V(B) \subseteq W$ since B meets V_k; hence B meets all of $V'_1, \ldots, V'_{k-1}$, and so B meets X''. But

$$|X' \cup X''| < 2^{n\theta^4} + \alpha(k-1, 2^{n\theta^4} + n + 1) = \alpha(k, n)$$

and so $X = X' \cup X''$ satisfies (ii), as required.

[p. 97] We remark that an assumption of the kind $G \in \mathcal{F}_\theta$ is needed in this result. For example, let G be the ϕ-grid with vertices v_{ij} $(1 \le i, j \le \phi)$; let

$$V_1 = \{v_{11}, \ldots, v_{\phi 1}\}$$
$$V_2 = \{v_{11}, \ldots, V_{1\phi}\}$$
$$V_3 = \{v_{1\phi}, \ldots, v_{\phi\phi}\}.$$

Take $n = 2$. Then (i) of (3.2) is false, as is easily seen; and the smallest set X making (ii) true has cardinality ϕ, which can be arbitrarily large.

4. Grids And Webs

Let G be a graph. An (m, n)-web in G is a pair $((A_1, \ldots, A_m), (B_1, \ldots, B_n))$, where $A_1, \ldots, A_m, B_1, \ldots, B_n$ are paths of G, such that

(i) $A_1, \ldots, A_m$ are disjoint, and $B_1, \ldots, B_n$ are disjoint;

(ii) each A_i meets each B_j;

(iii) $A_1, \ldots, A_m, B_1, \ldots, B_n$ are pairwise edge-disjoint.

(We permit $m = 0$ and $n = 0$.) The object of this section is to prove that if $G \in \mathcal{F}_\theta$ then G has no (θ_2, θ_2)-web, where θ_2 is as defined in Section 2.

(4.1) *Let $G \in \mathcal{F}_\theta$. Let $A_1, \ldots, A_m, B_1, \ldots, B_n$ be disjoint connected subgraphs of G such that for $1 \le i \le m$, $1 \le j \le n$ there is a vertex of A_i and a vertex of B_j which are adjacent in G. Then either $m < \theta^2/2$ or $n < \theta^2/2$.*

Proof. By contracting the edges of each of $A_1, \ldots, A_m, B_1, \ldots, B_n$ we obtain a graph which has a subgraph isomorphic to $K_{m,n}$. But $K_{\theta^2/2,\theta^2/2} \notin \mathcal{F}_\theta$, and the result follows.

(4.2) *Let $A_1, \ldots, A_{\theta^2/2}$ be disjoint connected subgraphs of $G \in \mathcal{F}_\theta$, and let $B_1, \ldots, B_n$ be pairwise edge-disjoint connected subgraphs of G. Suppose that for $1 \le i \le \theta^2/2$, $1 \le j \le n$ there is a unique vertex v_{ij} in $V(A_i) \cap V(B_j)$, and it has valency 1 in B_j. Suppose in addition that all vertices of G have valency ≤ 4. Then $n < \theta_1$, where θ_1 is as defined in Section 2.*

Proof. Put $m = \theta^2/2$. We may assume that each B_j is a tree with end-vertices $v_{1j}, \ldots, v_{mj}$. For $1 \le j \le n$, let B'_j be the tree obtained from B_j by deleting $v_{1j}, \ldots, v_{mj}$. Thus no vertex of G is in more than two of $B'_1, \ldots, B'_n$, since G has maximum valency ≤ 4. Let

$$G' = G \setminus (V(A_1) \cup \cdots \cup V(A_m)).$$

[p. 98] Then each B'_j is a subgraph of G'. For $1 \le i \le m$, $1 \le j \le n$, let u_{ij} be the (unique) vertex of B_j adjacent in B_j to v_{ij}. For $1 \le i \le m$, let $V_i = \{u_{ij} : 1 \le j \le n\}$, Then $V_1, \ldots, V_m \subseteq V(G')$ (since $m > 2$), and each B'_j meets each V_i.

By (4.1) there do not exist $\theta^2/2$ disjoint connected subgraphs of G' each meeting each of $V_1, \ldots, V_m$. Hence by (3.2) there exists $X \subseteq V(G')$ with $|X| < \alpha(\theta^2/2, \theta^2/2)$, such that every connected subgraph of G' which meets all of $V_1, \ldots, V_m$ also meets X. In particular, X meets each of $B_1, \ldots, B_n$. But any vertex of X is in at most two of $B_1, \ldots, B_n$, and so $|X| \ge n/2$. Thus $n/2 < \alpha(\theta^2/2, \theta^2/2)$ as required.

Let $((A_1, \ldots, A_m), (B_1, \ldots, B_n))$ be an (m, n)-web in a graph G. A spider of this web is a connected subgraph C of G such that

(i) C is edge-disjoint from $B_1, \ldots, B_n$;

(ii) for $1 \le i \le m$, C has a unique vertex in common with A_i, and that vertex has valency 1 in C.

We shall need the following, proved in [9, Theorem (2.8)].

(4.3) *Let $A_1, A_2 \subseteq V(G)$, where $G \in \mathcal{F}_\theta$, and let $D_1, \ldots, D_m$ be disjoint subgraphs of G, each with at most d components. Suppose that $d + n > 0$. Suppose that there are n disjoint paths in G from A_1 to A_2, and for $1 \le k \le m$, D_k separates A_1 and A_2. Then either $n < d\theta^2/2$ or $m <$*

$2^{d\theta^2/2}(\theta^2 + 2d\theta^{2\theta})$.[F3]

(4.4) *Let $G \in \mathcal{F}_\theta$, and let $p, q \geq 0$ be integers with $p > 0$.*[F4] *Let $m = p2^{p\theta^2}$, $n = p + q + m$. Suppose that $((A_1, \ldots, A_m), (B_1, \ldots, B_n))$ is an (m, n)-web in G. Then there exists $I \subseteq \{1, \ldots, m\}$, $J \subseteq \{1, \ldots, n\}$ such that $|I| = p$, $|J| = q$, and $((A_i : i \in I), (B_j : j \in J))$ is a (p, q)-web with a spider.*

Proof. We assume $p > 0$, for otherwise the result is trivial. We assume without loss of generality that every vertex and edge of G occurs in one of $A_1, \ldots, A_m, B_1, \ldots, B_n$. Let L be the simple graph with vertex set $\{1, \ldots, m\}$, in which distinct i, i' are adjacent if there exists $j \in \{1, \ldots, n\}$ such that a subpath of B_j has initial vertex in one of $A_i, A_{i'}$, terminal vertex in the other, and no interior vertex in any of $A_1, \ldots, A_m$. For each $X \subseteq V(L)$, we define $N(X)$ to be the set of vertices of L not in X but adjacent in L to a vertex of X.

Case 1. $|N(V(L'))| < p$ for every connected subgraph L' of L.

Define $Y_1 = \{1\}$, and inductively, having defined $Y_1, \ldots, Y_k$, define $Y_{k+1} = N(Y_1 \cup \cdots \cup Y_k)$. Then $|Y_k| < p$ for all $k > 0$. Define $m' = 2^{p\theta^2}$. Now L is connected and

$$|Y_1 \cup \cdots \cup Y_{m'-1}| \leq p(m' - 1) < m = |V(L)|,$$

[p. 99] and so $Y_{m'} \neq \emptyset$. We assume $2 \in Y_{m'}$ (say). Now for $1 \leq k \leq m'$, Y_k separates $\{1\}$ and $\{2\}$ in L, and so D_k separates A_1 and A_2 in G, where D_k is the subgraph of G with $|Y_k|$ components A_j $(j \in Y_k)$. But there are n disjoint paths in G from A_1 to A_2 (subpaths of $B_1, \ldots, B_n$) and so by (4.3), either $n < (p-1)\theta^2/2$ or

$$m' < 2^{(p-1)\theta^2/2}(\theta^2 + 2(p-1)\theta^{2\theta}).$$

But $n = p + q + p2^{p\theta^2}$ and $m' = 2^{p\theta^2}$, a contradiction.

Case 2. $|N(V(L'))| \geq p$ for some connected subgraph L' of L.

Choose $I \subseteq N(V(L'))$ with $|I| = p$. For each $i \in I$ choose $j_i \in \{1, \ldots, n\}$ such that there is a subpath of B_{j_i} joining A_i and A_h for some $h \in V(L')$, with no interior vertex in any of $A_1, \ldots, A_m$. Let P_i be such a subpath.

[F3] **Degenerate case.** For $d = n = 0$ the statement fails. Take $A_1 = A_2 = \emptyset$, $m = \theta^2$ and every D_k null. Then $n = 0 \not< 0$ and $m = \theta^2 \not< \theta^2$. The statement is (2.8) of Graph Minors I with $r = 0$ (and with m and n exchanged), which fails in the same way. The only use, in the proof of (4.4), has $n = p + q + m \geq p > 0$.

[F4] **Degenerate case.** For $p = q = 0$ and G null the statement fails. Then $m = n = 0$ and the empty families form a $(0, 0)$-web, but a spider is a connected subgraph of G, and the null graph has none. The paper allows null graphs (proof of (8.4), page 110). The only use, in the proof of (4.5), has $p = \phi_{k+1} \geq \theta^2/2 > 0$.

Choose a spanning tree of L' with edge-set E' say, and for each $e \in E'$ with ends i, i' say, choose $j_e \in \{1, \dots, n\}$ such that there is a subpath of B_{j_e} joining A_i and $A_{i'}$ with no interior vertex in any of $A_1, \dots, A_m$. Let Q_e be such a subpath. Let C be the subgraph of G consisting of the paths A_i ($i \in V(L')$), P_i ($i \in I$) and Q_e ($e \in E'$). Then C is connected, and has a unique vertex in common with each A_i ($i \in I$), and that vertex has valency 1 in C. Choose

$$J \subseteq \{1, \dots, n\} - \{j_i : i \in I\} - \{j_e : e \in E'\}$$

with $|J| = q$. This is possible since

$$n - |I| - |E'| \geq n - p - m = q.$$

Then $((A_i : i \in I), (B_j : j \in J))$ is a (p, q)-web with spider C. This completes the proof.

Define $\phi_{\theta_1}, \dots, \phi_0$ as in Section 2. Define $\psi_{\theta_1} = 0$, and for $0 \leq k < \theta_1$, define

$$\psi_k = \phi_k + 2\phi_{k+1} + 2\phi_{k+2} + \cdots + 2\phi_{\theta_1 - 1} + \phi_{\theta_1}.$$

Then for $0 \leq k < \theta_1$ we have $\psi_k = \psi_{k+1} + \phi_k + \phi_{k+1}$.

(4.5) *For $0 \leq k < \theta_1$,*[F5] *if $G \in \mathcal{F}_\theta$ has a (ϕ_k, ψ_k)-web with k edge-disjoint spiders, then it has a (ϕ_{k+1}, ψ_{k+1})-web with $k+1$ edge-disjoint spiders.*

Proof. Let $((A_1, \dots, A_{\phi_k}), (B_1, \dots, B_{\psi_k}))$ be a (ϕ_k, ψ_k)-web with k edge-disjoint spiders $C_1, \dots, C_k$. Let G' be obtained from G by deleting the edges of $C_1, \dots, C_k$. Then $((A_1, \dots, A_{\phi_k}), (B_1, \dots, B_{\psi_k}))$ is a (ϕ_k, ψ_k)-web in G'. By (4.4) there exists $I \subseteq \{1, \dots, \phi_k\}$ with $|I| = \phi_{k+1}$, and $J \subseteq \{1, \dots, \psi_k\}$ with $|J| = \psi_{k+1}$ such that $((A_i : i \in I), (B_j : j \in J))$ is a (ϕ_{k+1}, ψ_{k+1})-web in G' with a spider C_{k+1}. But then this is a web in G, and in G it has $k+1$ edge-disjoint spiders $C_1, \dots, C_{k+1}$.

[p. 100] (4.6) *If $G \in \mathcal{F}_\theta$ then G has no (θ_2, θ_2)-web.*

Proof. Suppose $G \in \mathcal{F}_\theta$ and that G has a (θ_2, θ_2)-web. Delete the vertices and edges of G not used by any paths in the web, forming G'. Then $G' \in \mathcal{F}_\theta$, and all vertices of G' have valency ≤ 4. Moreover G' has a (θ_2, θ_2)-web and hence a (ϕ_0, ψ_0)-web. By (4.5) it has a $(\phi_{\theta_1}, \psi_{\theta_1})$-web with θ_1 edge-disjoint spiders. But $\phi_{\theta_1} = \theta^2/2$, contradicting (4.2).

5. Webs and Meshes

[F5] **Misprint.** Here ϕ_k and ψ_k are defined only for $0 \leq k \leq \theta_1$ (three lines above), and (4.6) applies (4.5) up to $k = \theta_1$.

An (m,n)-mesh in a graph G is a pair $((A_1,...,A_m),(B_1,...,B_n))$, where
(i) $A_1,...,A_m,B_1,...,B_n$ are all connected subgraphs of G;
(ii) $A_1,...,A_m$ are disjoint, and $B_1,...,B_n$ are disjoint;
(iii) each A_i meets each B_j.

The differences between meshes and webs are first that in meshes the A_i's and B_j's are arbitrary connected subgraphs instead of paths, as in webs, and second, that we now permit A_i and B_j to have common edges. The object of this section is to prove that if $G \in \mathcal{F}_\theta$ then G has no (θ_5,θ_6)-mesh, where θ_5,θ_6 are as defined in Section 2. We begin with a lemma.

(5.1) *Let v be a vertex of a connected graph G, and let $p,q \geq 0$ be integers with $p \geq 2$.*[F6] *Let $V_1,...,V_{p^q} \subseteq V(G)$ be disjoint and non-empty, with $v \in V_1 \cup \cdots \cup V_{p^q}$. Then at least one of the following holds:*

(i) *there is a connected subgraph C of G with $v \in V(C)$, and there are p vertices $v_1,...,v_p \neq v$ of C, each with valency 1 in C, and there are distinct $j_1,...,j_p \in \{1,...,p^q\}$, such that*

$$V_{j_i} \cap V(C) = \{v_i\} \qquad (1 \leq i \leq p);$$

(ii) *there is a path of G with initial vertex v, meeting exactly $q+1$ of $V_1,...,V_{p^q}$.*

Proof. We proceed by induction on q. The result is trivial if $q = 0$, and we assume $q > 0$. Let $v \in V_{p^q}$, say. Choose a maximal connected subgraph L of G with $v \in V(L)$ such that

$$V(L) \cap V_j = \emptyset \qquad (1 \leq j < p^q).$$

Let $B_1,...,B_n$ be the components of $G \setminus V(L)$. For $1 \leq k \leq n$, let J_k be the set of those $j \in \{1,...,p^q-1\}$ such that $V_j \cap V(B_k) \neq \emptyset$.

Case 1. $|J_k| < p^{q-1}$ for all k $(1 \leq k \leq n)$.

[p. 101] Now $J_1 \cup \cdots \cup J_n = \{1,...,p^q-1\}$, since each V_j is non-empty. Choose $N \subseteq \{1,...,n\}$ minimal such that

$$\bigcup_{k \in N} J_k = \{1,...,p^q-1\}.$$

Then

$$|N| \geq \frac{p^q-1}{p^{q-1}-1} \geq p$$

[F6] **Degenerate case.** For $p = 1$ the statement fails. Let G have one vertex v, let $p = q = 1$, and let $V_1 = \{v\}$. Then (i) needs a vertex $v_1 \neq v$ in $V_1 \cap V(C)$, and (ii) needs a path meeting $q+1 = 2$ of the single set V_1. The only use, in the proof of (5.2) on page 102, has $p = \theta^2/2 \geq 18$.

and by the minimality of N, for each $k \in N$ there exists $j(k) \in \{1, ..., p^q - 1\}$ such that $j(k) \in J_k$ and such that $j(k) \notin J_{k'}$ for each $k' \in N - \{k\}$. Let L' be the subgraph of G induced on the set

$$V(L) \cup \bigcup (V(B_k) : k \in N).$$

Then L' is connected, and meets V_j for all $j \in \{1, ..., p^q - 1\}$, and $v \in V(L')$. Let $j_1, ..., j_p$ be the numbers $j(k)$ corresponding to p distinct numbers $k \in N$. Let C be a minimal connected subgraph of L' with $v \in V(C)$ such that $V(C) \cap V_{j_i} \neq \emptyset$ $(1 \leq i \leq p)$. Then C satisfies (i), as may be easily seen.

Case 2. $|J_k| \geq p^{q-1}$ for some k $(1 \leq k \leq n)$.

Let u be a vertex of B_k adjacent in G to a vertex of L. By the maximality of L, $u \in V_i$ for some $i \in \{1, ..., p^q - 1\}$. Choose $J \subseteq J_k$ with $i \in J$ and $|J| = p^{q-1}$. We apply our inductive hypothesis to B_k, u, $V_j \cap V(B_k)$ $(j \in J)$. If (i) holds here then it holds for G, and if (ii) holds here then it holds for G. This completes the proof.

(5.2) *Let $G \in \mathcal{F}_\theta$, and let $p, q \geq 2$ be integers. Let $n = (\theta^2/2)^{q-1}$, $m = p\binom{n}{q} + \frac{1}{2}\theta^2\binom{n}{\theta^2/2}$. Suppose that $((A_1, ..., A_m), (B_1, ..., B_n))$ is an (m, n)-mesh in G. Then there exists $I \subseteq \{1, ..., m\}$ and $J \subseteq \{1, ..., n\}$ with $|I| = p$, $|J| = q$, and for each $i \in I$ a path P_i of A_i, such that $((P_i : i \in I), (B_j : j \in J))$ is a (p, q)-mesh.*

Proof. Fix i with $1 \leq i \leq m$ for the moment. For $1 \leq j \leq n$, let $V_j = V(A_i) \cap V(B_j)$. Then $V_1, ..., V_n$ are disjoint non-empty subsets of $V(A_i)$. Let us say that i is *starred* if there is a connected subgraph C of A_i and there are vertices $v_1, ..., v_{\theta^2/2}$ of C, each with valency 1 in C, and there are distinct $j_1, ..., j_{\theta^2/2} \in \{1, ..., n\}$, such that

$$V_{j_k} \cap V(C) = \{v_k\} \qquad (1 \leq k \leq \theta^2/2).$$

[p. 102] If i is starred, let $J(i)$ be the set $\{j_1, ..., j_{\theta^2/2}\}$. Let the set of all starred $i \in \{1, ..., m\}$ be I^*, and let $|I^*| = N$. Then there exists $I \subseteq I^*$ with

$$|I| \geq \binom{n}{\theta^2/2}^{-1} \cdot N$$

such that $J(i)$ is the same for all $i \in I$. Let $J(i) = J$ say $(i \in I)$. Now $|J| = \theta^2/2$, and so by (4.1), $|I| < \theta^2/2$. Hence

$$N < \frac{1}{2}\theta^2 \binom{n}{\theta^2/2}.$$

It follows that there exists $I' \subseteq \{1, ..., m\} - I^*$ with

$$|I'| = m - \frac{1}{2}\theta^2 \binom{n}{\theta^2/2} = p\binom{n}{q}.$$

Now each $i \in I'$ is not starred, and so by (5.1) there is a path P_i of A_i meeting exactly q of $B_1, ..., B_n$. There exists $I'' \subseteq I'$ with

$$|I''| = \binom{n}{q}^{-1} \cdot |I'| = p$$

and $J'' \subseteq \{1, ..., n\}$ with $|J''| = q$, such that P_i meets B_j for $i \in I''$, $j \in J''$. This is the required mesh.

The result stated earlier as (1.6) is now readily proved as follows.

(5.3) *If $G \in \mathcal{F}_\theta$ then G has no (θ_5, θ_6)-mesh.*

Proof. If G has a (θ_5, θ_6)-mesh then it has a (θ_6, θ_5)-mesh, and so by (5.2) it has a (θ_3, θ_4)-mesh $((A_1, ..., A_{\theta_3}), (B_1, ..., B_{\theta_4}))$, where each A_i is a path. We apply (5.2) to the mesh $((B_1, ..., B_{\theta_4}), (A_1, ..., A_{\theta_3}))$, and deduce that G has a (θ_2, θ_2)-mesh $((C_1, ..., C_{\theta_2}), (D_1, ..., D_{\theta_2}))$, where each C_i and each D_j is a path. We contract every edge of G which is both in some C_i and in some D_j, and obtain a graph $G' \in \mathcal{F}_\theta$ which has a (θ_2, θ_2)-web, contrary to (4.6).

6. Large Subgraphs and Small Separations

It remains to show that if a graph has no large mesh then its tree-width is bounded. This will be proved in the next section. This section is devoted to a lemma which will be used in the proof. The following result is due to Tutte, implicitly in [13].

[p. 103] (6.1) *If G is a 2-connected graph with n vertices, we may enumerate its vertices as $v_1, ..., v_n$ in such a way that for $0 \le i \le n$, $G|\{v_1, ..., v_i\}$, $G|\{v_{i+1}, ..., v_n\}$ are both connected.*

Let $\mathbf{Z}^+$ denote the set of non-negative integers. If G is a graph and $w\colon V(G) \to \mathbf{Z}^+$ is a function and $X \subseteq V(G)$, then $w(X)$ denotes $\sum_{v \in X} w(v)$. We use (6.1) to prove the following.

(6.2) *Let G be a connected graph, let $w\colon V(G) \to \mathbf{Z}^+$ be some function, and let $k > 0$ be an integer with $k \le (w(V(G)) + 2)/3$. Then exactly one of the following is true:*

(i) *there are disjoint subsets V_1, V_2 of $V(G)$ such that $G|V_1$, $G|V_2$ are both connected and $w(V_1), w(V_2) \ge k$;*

(ii) *there is a vertex v of G such that $w(V(C)) < k$ for every component C of $G \setminus v$.*

Proof. Clearly not both (i) and (ii) hold. We prove that at least one of them holds by induction on $|V(G)|$. Suppose that G is not 2-connected, and let u be a vertex of G such that $G \setminus u$ has more than one component. Let $V_1, ..., V_r$ be the vertex sets of the components of $G \setminus u$. If $w(V_1), ..., w(V_r) < k$ then (ii) is true. We assume then that $w(V_1) \geq k$. Define $w_1 \colon \{u\} \cup V_1 \to \mathbf{Z}^+$ by

$$w_1(v) = w(v) \qquad (v \in V_1)$$
$$w_1(u) = w(\{u\} \cup V_2 \cup \cdots \cup V_r).$$

Then $w_1(\{u\} \cup V_1) = w(V(G))$. Let G_1 be $G|(\{u\} \cup V_1)$. Now G_1 has fewer vertices than G, and so by induction the result holds for G_1, w_1.

If (i) holds for G_1, w_1 then it holds for G, w, as is easily seen. We assume then that (ii) holds for G_1, w_1, and that v is a vertex of G_1 such that $w_1(V(C)) < k$ for every component C of $G_1 \setminus v$. Now $v \neq u$, because $G_1 \setminus u$ has only one component, and its vertex set is V_1, and $w_1(V_1) = w(V_1) \geq k$. Hence $w(V(C)) < k$ for every component C of $G \setminus v$, and (ii) is true.

We may assume therefore that G is 2-connected, so that (6.1) applies. Let G have vertices $v_1, ..., v_n$, enumerated so that for $0 \leq i \leq n$, $G|\{v_1, ..., v_i\}$, $G|\{v_{i+1}, ..., v_n\}$ are both connected. Choose an integer i with $0 \leq i \leq n$, minimum such that

$$\sum(w(v_j) : 1 \leq j \leq i) \geq k.$$

If

$$\sum(w(v_j) : i < j \leq n) \geq k$$

[p. 104] then (i) holds, and so we assume this is false, that is,

$$\sum(w(v_j) \colon 1 \leq j \leq i) \geq w(V(G)) - k + 1.$$

By the minimality of i,

$$\sum(w(v_j) \colon 1 \leq j < i) \leq k - 1$$

and so

$$w(v_i) \geq w(V(G)) - 2k + 2 \geq k$$

since $k \leq (w(V(G)) + 2)/3$. Now $G \setminus v_i$ is connected, since G is 2-connected, and so we may assume that $w(V(G) - \{v_i\}) < k$, for otherwise (i) is true. But in that case (ii) is true, as required.

We use (6.2) to prove the following, the main result of this section. (For a related theorem, see [7, 8].)

(6.3) *Let G be a connected graph, and let $k > 0$ be an integer. Then at least one of the following is true:*

(i) *there are k disjoint connected subgraphs of G, each with at least $2(3^k - 1)^{-1}|V(G)|$ vertices;*

(ii) *there exists $X \subseteq V(G)$ with $|X| < k$ such that every component of $G \setminus X$ has fewer than $|V(G)|/3$ vertices.*

Proof. If $\mathcal{B}$ is a set of k disjoint connected subgraphs of G, we define $d(\mathcal{B})$ to be the k-tuple $(b_1, ..., b_k)$, where the members of $\mathcal{B}$ have vertex sets of cardinality $b_1, ..., b_k$, respectively, and $b_1 \leq b_2 \leq \cdots \leq b_k$. Choose such a set $\mathcal{B}$ with $d(\mathcal{B})$ lexicographically last; that is, there is no set $\mathcal{B}'$ with $d(\mathcal{B}') = (b'_1, ..., b'_k)$ say, where for some j $(1 \leq j \leq k)$, $b'_j > b_j$ and $b'_i = b_i$ for $1 \leq i < j$. If $b_1 \geq 2(3^k - 1)^{-1}|V(G)|$ then (i) is true, and so we assume that $b_1 < 2(3^k - 1)^{-1}|V(G)|$ and shall prove that (ii) is true. Let $\mathcal{B} = \{B_1, ..., B_k\}$, where $V(B_i) = V_i$ and $|V_i| = b_i$ $(1 \leq i \leq k)$. From the lexicographic lastness of $d(\mathcal{B})$,

$$V_1 \cup \cdots \cup V_k = V(G)$$

since G is connected; and so

$$\sum_{1 \leq i \leq k} b_i = |V(G)|.$$

Now

$$\sum_{1 \leq i \leq k} 3^{i-1} b_1 < |V(G)|$$

[p. 105] by our assumption, and so $b_i > 3^{i-1}b_1$, for some i $(1 \leq i \leq k)$. Choose i' minimum such that $b_{i'} > 3^{i'-1}b_1$. Then $1 < i' \leq k$, and $b_{i'-1} \leq 3^{i'-2}b_1$ by the minimality of i'. Hence $b_{i'} > 3b_{i'-1}$.

Now for $i' \leq i \leq k$, $b_i \geq b_{i'} > 3^{i'-1}b_1 \geq 3b_1$. But B_i is a connected graph, and it does not have two disjoint connected subgraphs both with at least $b_1 + 1$ vertices, from the lexicographic lastness of $d(\mathcal{B})$. Hence by (6.2) with $w \equiv 1$, there is a vertex v_i of B_i such that every component of $B_i \setminus v_i$ has at most b_1 vertices. Let $X = \{v_{i'}, ..., v_k\}$. We shall show that X satisfies (ii). Let C be a component of $G \setminus X$, and suppose for a contradiction that $|V(C)| \geq |V(G)|/3$. There are two cases.

Case 1. $V(C) \cap (V_1 \cup \cdots \cup V_{i'-1}) \neq \emptyset$.

Now

$$|V_1 \cup \cdots \cup V_{i'-1}| \leq b_1 + 3b_1 + \cdots + 3^{i'-2}b_1$$

by the minimality of i', and so

$$|V_1 \cup \cdots \cup V_{i'-1}| \leq (3^{i'-1} - 1)b_1/2 \leq (3^k - 1)b_1/6 < |V(G)|/3.$$

Hence $V(C) \cap (V_{i'} \cup \cdots \cup V_k) \neq \emptyset$. But C is connected, $V(C) \cap X = \emptyset$ and $V(G) = V_1 \cup \cdots \cup V_k$, and so there exist $u \in V_1 \cup \cdots \cup V_{i'-1}$, $u' \in V_{i'} \cup \cdots \cup V_k$ which are adjacent vertices of C, and not in X. Let $u \in V_j$, $u' \in V_{j'}$ say, where $1 \leq j \leq i' - 1$, $i' \leq j' \leq k$. Let B be the component of $B_{j'} \setminus v_{j'}$ which contains u'. Let $B'_j = G|(V(B_j) \cup V(B))$. Then

$$|V(B_{j'} \setminus V(B))| = b_{j'} - |V(B)| > 3b_{i'-1} - b_1 \geq b_j,$$

and $B_{j'} \setminus V(B)$, B'_j are both connected; thus the set

$$\{B_{j'} \setminus V(B), B'_j\} \cup \{B_i \colon 1 \leq i \leq k, i \neq j, j'\}$$

contradicts the lexicographic lastness of $d(\mathcal{B})$.

Case 2. $V(C) \cap (V_1 \cup \cdots \cup V_{i'-1}) = \emptyset$.

Hence $V(C) \subseteq (V_{i'} \cup \cdots \cup V_k) - X$. Let us say that $V' \subseteq V(G)$ is a *piece* if for some i ($i' \leq i \leq k$), V' is the vertex set of a component of $B_i \setminus v_i$. Any two pieces are thus disjoint. Now $V(C)$ is a union of pieces, and

$$|V(C)| \geq |V(G)|/3 > (3^k - 1)b_1/6 \geq b_1$$

(unless $k = 1$, when the theorem is trivial). Hence there are pieces $P_1, ..., P_t$ say, such that $G|(P_1 \cup \cdots \cup P_t)$ is connected, $|P_1 \cup \cdots \cup P_t| > b_1$, and $|P_1 \cup \cdots \cup P_{t-1}| \leq b_1$. Now P_t is a piece and so $|P_t| \leq b_1$; hence [p. 106] $|P_1 \cup \cdots \cup P_t| \leq 2b_1$. Put $P = P_1 \cup \cdots \cup P_t$. Then $|V_{i'} - P|, ..., |V_k - P| > b_1$, and $B_{i'} \setminus P, ..., B_k \setminus P$ are all connected; and so the set

$$\{G|P, B_2, ..., B_{i'-1}, B_{i'} \setminus P, ..., B_k \setminus P\}$$

contradicts the lexicographic lastness of $d(\mathcal{B})$.

Since the assumption $|V(C)| \geq |V(G)|/3$ leads to a contradiction in both cases 1 and 2, every component of $G \setminus X$ has fewer than $|V(G)|/3$ vertices. Moreover $|X| < k$ since $i' > 1$. This completes the proof.

The following extends (6.3) to disconnected graphs.

(6.4) *Let G be a graph and let $k > 0$ be an integer. Then at least one of the following holds:*

(i) *there are k disjoint connected subgraphs of G, each with at least $4(3^k - 1)^{-1}|V(G)|/3$ vertices;*

(ii) *there is a separation* (V_1, V_2) *of* G *such that* $|V_1 \cap V_2| < k$ *and* $|V_1 - V_2|, |V_2 - V_1| \leq 2|V(G)|/3$.

Proof. Let G have components $G_1, ..., G_t$, where $|V(G_1)| \geq |V(G_2)| \geq \cdots \geq |V(G_t)|$. If $|V(G_1)| \leq 2|V(G)|/3$ then (ii) is true; indeed, we may choose V_1, V_2 in (ii) with $V_1 \cap V_2 = \emptyset$, as is easily seen. We assume then that $|V(G_1)| > 2|V(G)|/3$, and apply (6.3) to G_1. If (6.3) (i) holds for G_1, then (i) holds for G. We assume then that (6.3) (ii) holds for G_1, and $X \subseteq V(G_1)$ is such that $|X| < k$ and every component of $G_1 \setminus X$ has fewer than $|V(G_1)|/3$ vertices. Now

$$|V(G_2)|, ..., |V(G_t)| \leq |V(G)| - |V(G_1)| < |V(G)|/3,$$

and so every component of $G \setminus X$ has fewer than $|V(G)|/3$ vertices. It follows easily that (ii) holds, as required.

7. Meshes and Tree-Width

In this section we complete the proof of (2.1).

(7.1) *If* $G \in \mathcal{F}_\theta$ *then* G *has a separation* (V_1, V_2) *such that* $|V_1 \cap V_2| \leq \theta_7$ *and* $|V_1 - V_2|, |V_2 - V_1| \leq (1 - \theta_8^{-1})|V(G)|$, *where* θ_7, θ_8 *are as defined in Section 2.*

Proof. If there is a separation (V_1, V_2) of G with $|V_1 \cap V_2| < \theta_5$, and

$$|V_1 - V_2|, |V_2 - V_1| \leq 2|V(G)|/3$$

[p. 107] then the result is true, since $\theta_5 \leq \theta_7$ and $2/3 \leq 1 - \theta_8^{-1}$. We assume then there is no such separation, so that by (6.4) there are θ_5 disjoint connected subgraphs $A_1, ..., A_{\theta_5}$ of G each with at least

$$4(3^{\theta_5} - 1)^{-1}|V(G)|/3 = \theta_5\theta_8^{-1}|V(G)|$$

vertices. Now by (5.3), G has no (θ_5, θ_6)-mesh, and so there do not exist θ_6 disjoint connected subgraphs each meeting all of $A_1, ..., A_{\theta_5}$. By (3.2), there exists $X \subseteq V(G)$ with $|X| < \alpha(\theta_5, \theta_6) = \theta_7$ such that every connected subgraph of G which meets all of $A_1, ..., A_{\theta_5}$ also meets X. Thus, for each $v \in V(A_1)$, X separates $\{v\}$ from one of $A_2, ..., A_{\theta_5}$, say $A_{i(v)}$.

Choose $V' \subseteq V(A_1)$ with $|V'| \geq \theta_5^{-1}|V(A_1)|$ such that $i(v)$ is the same for all $v \in V'$. Let $i(v) = 2$ say, for all $v \in V'$. Then

$$|V'| \geq \theta_5^{-1}|V(A_1)| \geq \theta_8^{-1}|V(G)|.$$

Choose a separation (V_1, V_2) of G with $V_1 \cap V_2 = X$, $V' \subseteq V_1$ and $V(A_2) \subseteq V_2$. Then $|V_1 \cap V_2| \leq \theta_7$, and

$$|V_1 - V_2| \leq |V(G)| - |V(A_2)| \leq (1 - \theta_8^{-1})|V(G)|$$
$$|V_2 - V_1| \leq |V(G)| - |V'| \leq (1 - \theta_8^{-1})|V(G)|.$$

This completes the proof.

(7.2) *If $G \in \mathcal{F}_\theta$ and $X \subseteq V(G)$ then G has a separation (V_1, V_2) such that $|V_1 \cap V_2| \leq \theta_7$ and $|(V_1 - V_2) \cap X|, |(V_2 - V_1) \cap X| \leq (1 - \theta_8^{-1})|X|$.*

Proof. Let N be an integer so large that there is no integer greater than $(1 - \theta_8^{-1})|X|$ and not greater than

$$(1 - \theta_8^{-1})|X| + N^{-1}((1 - \theta_8^{-1})|V(G)| + \theta_7).$$

For each $v \in X$, let N_v be a set of N new vertices, and let G' be the graph with vertex set

$$V(G) \cup \bigcup_{v \in X} N_v$$

such that $G' \setminus \bigcup_{v \in X} N_v = G$, and for each $v \in X$, every $u \in N_v$ is adjacent in G' to v and to no other vertex. Evidently $G' \in \mathcal{F}_\theta$, and so by (7.1) there is a separation (V_1', V_2') of G' with $|V_1' \cap V_2'| \leq \theta_7$ and

$$|V_1' - V_2'|, |V_2' - V_1'| \leq (1 - \theta_8^{-1})|V(G')|.$$

[p. 108] Let $V_1 = V_1' \cap V(G)$, $V_2 = V_2' \cap V(G)$. Now if $v \in (V_1 - V_2) \cap X$ then $N_v \subseteq V_1'$, and so

$$\begin{aligned} N \cdot |(V_1 - V_2) \cap X| &\leq |V_1'| \\ &\leq (1 - \theta_8^{-1})|V(G')| + \theta_7 \\ &\leq (1 - \theta_8^{-1})(N|X| + |V(G)|) + \theta_7. \end{aligned}$$

Hence

$$|(V_1 - V_2) \cap X| \leq (1 - \theta_8^{-1})|X| + N^{-1}((1 - \theta_8^{-1})|V(G)| + \theta_7)$$

and so by the choice of N,

$$|(V_1 - V_2) \cap X| \leq (1 - \theta_8^{-1})|X|.$$

Similarly

$$|(V_2 - V_1) \cap X| \leq (1 - \theta_8^{-1})|X|.$$

But $|V_1 \cap V_2| \leq |V_1' \cap V_2'| \leq \theta_7$, and (V_1, V_2) is a separation of G. This completes the proof.

We come to the principal theorem of this paper.

(7.3) *If $G \in \mathcal{F}_\theta$ then G has tree-width at most θ_9.*

Proof. We show, by induction on $|V(G)|$, that if $G \in \mathcal{F}_\theta$ and $X \subseteq V(G)$ and $|X| \leq \theta_7\theta_8 + 1$, then G has a tree-decomposition $(T, (X_t : t \in V(T)))$ of width $\leq \theta_9$, such that $X \subseteq X_t$, for some $t \in V(T)$. We assume the result holds for all graphs with fewer vertices than G. We may suppose that $|V(G)| > \theta_9$, for otherwise the result is clear. Hence we may assume that $|X| = \theta_7\theta_8+1$. By (7.2), there is a separation (V_1, V_2) of G with $|V_1 \cap V_2| \leq \theta_7$ and

$$|(V_1 - V_2) \cap X|, |(V_2 - V_1) \cap X| \leq (1 - \theta_8^{-1})|X|.$$

But

$$|V_1 \cap V_2 \cap X| \leq |V_1 \cap V_2| \leq \theta_7$$

and so

$$\begin{aligned} |V_1 \cap X| &= |(V_1 - V_2) \cap X| + |V_1 \cap V_2 \cap X| \\ &\leq (1 - \theta_8^{-1})|X| + \theta_7 < |X|. \end{aligned}$$

Hence $V_1 \neq V(G)$. Put $G_1 = G|V_1$, and

$$X_1 = ((V_1 - V_2) \cap X) \cup (V_1 \cap V_2).$$

[p. 109] Now $G_1 \in \mathcal{F}_\theta$, and has fewer vertices than G, and

$$|X_1| \leq (1 - \theta_8^{-1})|X| + \theta_7 \leq \theta_7\theta_8 + 1.$$

Thus by induction G_1 has a tree-decomposition $(T_1, (X_t : t \in V(T_1)))$ of width $\leq \theta_9$, such that $X_1 \subseteq X_{t_1}$ for some $t_1 \in V(T_1)$. Define G_2, X_2, T_2, t_2 and a tree-decomposition $(T_2, ((X_t : t \in V(T_2))))$ similarly for $G|V_2$. We assume without loss of generality that T_1, T_2 are disjoint. Take a new vertex $t_0 \notin V(T_1) \cup V(T_2)$, and let T be the tree with vertex set $\{t_0\} \cup V(T_1) \cup V(T_2)$, in which t_0 is adjacent to t_1 and t_2, and $T \setminus t_0$ has two components T_1, T_2. Define $X_{t_0} = X \cup (V_1 \cap V_2)$. Then $|X_{t_0}| \leq \theta_9$, and it is easy to see that $(T, (X_t : t \in V(T)))$ is the required tree-decomposition of G. This completes the inductive argument, and the theorem follows.

8. A Generalization of the Erdös-Pósa Theorem

The following result is due to Erdös and Pósa [4].

(8.1) *For any integer $k \geq 0$ there is an integer $k' \geq 0$ such that for every graph G, at least one of the following is true:* (i) *G has k disjoint circuits;* (ii) *there exists $X \subseteq V(G)$ with $|X| \leq k'$ such that $G \setminus X$ has no circuits.*

It is natural to ask if a similar result holds for other structures instead of circuits. A circuit of G may be regarded as a subgraph of G contractible to a loop. Motivated by this, we say that a subgraph G' of a graph G is an *H-expansion in G* (where H is a graph) if some edges of G' may be contracted to produce a graph isomorphic to H. For example, if H is the graph consisting of a single loop, the H-expansions in a graph G are precisely those connected subgraphs of G which have circuits. Thus, the maximum number of disjoint H-expansions equals the maximum number of disjoint circuits; and $X \subseteq V(G)$ meets every H-expansion if and only if it meets every circuit.

Let us say that a graph H has the *Erdös-Pósa property* if for every integer $k \geq 0$ there is an integer $k' \geq 0$ such that for every graph G, at least one of the following holds: (i) there are k pairwise vertex-disjoint H-expansions in G (ii) there exists $X \subseteq V(G)$ with $|X| \leq k'$ such that X meets every H-expansion in G.

Then (8.1) asserts that the graph consisting of a single loop has the Erdös-Pósa property. We have the following curious characterization.

[p. 110] (8.2) *A graph has the Erdös-Pósa property if and only if it is planar.*

The following lemmas are used to derive (8.2) from our main result (1.5). We say that a simple graph G is a *subtree intersection graph* if there is a tree T and a family $(T_v : v \in V(G))$ of subtrees of T such that for distinct $v, v' \in V(G)$, v, v' are adjacent in G if and only if T_v and $T_{v'}$ have a common vertex. We say that a simple graph G is *chordal* if it has no induced subgraph which is a circuit of four or more vertices. The following is shown in [2, 5].

(8.3) *G is a subtree intersection graph if and only if G is chordal.*

(In fact for our application we need only the easier "only if" direction.) The following is due to Dirac [3].

(8.4) *If G is chordal and non-null then G has a vertex v such that the vertices adjacent to v are pairwise adjacent.*

Our main lemma is the following.

(8.5) *Let G be chordal, and let $Z_1, \ldots, Z_m$ be independent subsets of $V(G)$, pairwise disjoint, and each with cardinality k. Let $x_1, \ldots, x_m \geq 0$ be integers with $x_1 + \cdots + x_m = k$. Then there exists an independent subset $X \subseteq V(G)$ with $|X \cap Z_i| = x_i$ $(1 \leq i \leq m)$.*

Proof. We proceed by induction on $|V(G)|$; if G is null the result is true, and so we assume that G is non-null, and that the theorem holds for all graphs with fewer vertices. If there is a vertex $u \in V(G)$ with $u \notin Z_1 \cup \cdots \cup Z_m$ the result follows by our inductive hypothesis applied to $G \setminus u$, which is chordal. We assume then that

$$Z_1 \cup \cdots \cup Z_m = V(G).$$

By (8.4), G has a vertex v such that all vertices of G adjacent to v are pairwise adjacent. Now $v \in Z_1 \cup \cdots \cup Z_m$; we assume that $v \in Z_m$ without loss of generality. If $x_m = 0$ the result follows from our inductive hypothesis applied to $G \setminus v$ with $m-1$ independent subsets $Z_1, \ldots, Z_{m-1}$. We assume then that $x_m \neq 0$. For $1 \leq i \leq m-1$, choose $v_i \in Z_i$ adjacent to v if possible, and otherwise arbitrarily.

Every neighbor u of v in G is one of $v_1, \ldots, v_{m-1}$; for if u is adjacent to v, then $u \notin Z_m$ (since Z_m is independent) and so $u \in Z_i$ for some i with $1 \leq i < m$; hence v_i is adjacent to v, but not to u since Z_i is independent, and so $v_i = u$. Define $x'_1 = x_1, \ldots, x'_{m-1} = x_{m-1}$, $x'_m = x_m - 1$, $G' = G \setminus \{v_1, \ldots, v_{m-1}, v\}$, $k' = k-1$, $Z'_i = Z_i - \{v_i\}$ $(1 \leq i \leq m-1)$, $Z'_m = Z_m - \{v\}$. Then $Z'_1, \ldots, Z'_m$ are disjoint independent subsets of $V(G')$, each of [p. 111] cardinality k', and so by induction there is an independent subset $X' \subseteq V(G')$ with $|X' \cap Z'_i| = x'_i$ $(1 \leq i \leq m)$. Then $X = X' \cup \{v\}$ is independent in G, and satisfies the theorem, as required.

If we apply the foregoing to trees we obtain the following.

(8.6) *Let T be a tree, and let $\mathcal{A}_1, \ldots, \mathcal{A}_m$ be families of subtrees of T. Let $x_1, \ldots, x_m \geq 0$ be integers, and let $x_1 + \cdots + x_m = k$. Suppose that for $1 \leq i \leq m$, there are k members of $\mathcal{A}_i$ which are pairwise vertex-disjoint. Then for $1 \leq i \leq m$ there are x_i members $T^i_1, \ldots, T^i_{x_i}$ of $\mathcal{A}_i$, such that*

$$T^1_1, \ldots, T^1_{x_1}, T^2_1, \ldots, T^2_{x_2}, \ldots, T^m_1, \ldots, T^m_{x_m}$$

are all pairwise vertex-disjoint.

Proof. For $1 \leq i \leq m$, let $S^i_1, \ldots, S^i_k$ be pairwise vertex-disjoint members of $\mathcal{A}_i$. Let G be the subtree intersection graph associated with $S^1_1, \ldots, S^1_k, S^2_1, \ldots, S^2_k, \ldots, S^m_1, \ldots, S^m_k$. The theorem follows by applying (8.5) to G, which is chordal by (8.3).

We also require the following, which is a standard result in hypergraph theory; see, for example, [6].

(8.7) *Let T be a tree, and let $\mathcal{F}$ be a family of subtrees of T. Then for any integer $k \geq 0$, either there are k pairwise vertex-disjoint members of $\mathcal{F}$, or there is a subset $X \subseteq V(T)$ with $|X| < k$ which has non-empty intersection with every member of $\mathcal{F}$.*

We can now prove one half of (8.2).

(8.8) *If H is planar then H has the Erdös-Pósa property.*

Proof. Let k be a positive integer, and let H' be the graph consisting of k disjoint copies of H. Then H' is planar, and so by (1.5) there is a number $w \geq 0$ such that every graph with no minor isomorphic to H' has tree-width $\leq w$. Let H have m connected components, and put $k' = (mk-1)(w+1)$. We shall show that k' satisfies condition (ii) of the Erdös-Pósa property. Let the components of H be $H_1, \ldots, H_m$.

Suppose then that G is a graph, and there do not exist k vertex-disjoint H-expansions in G. Then G has no minor isomorphic to H', and so G has tree-width $\leq w$. Let $(T, (X_t : t \in V(T)))$ be a tree-decomposition of G such that $|X_t| \leq w+1$ for each $t \in V(T)$. For $1 \leq j \leq m$, let $\mathcal{A}_j$ be the set of all H_j-expansions in G, and for each $A \in \mathcal{A}_j$ let

$$T(A) = \{t \in V(T) : V(A) \cap X_t \neq \emptyset\}.$$

[p. 112] Now A is connected, since it is an H_j-expansion and H_j is connected, and so $T(A)$ is the vertex set of a subtree of T. There do not exist

$$\begin{gathered} A_{11}, \ldots, A_{1k} \in \mathcal{A}_1 \\ \vdots \qquad\qquad \vdots \\ A_{m1}, \ldots, A_{mk} \in \mathcal{A}_m \end{gathered}$$

such that $T(A_{11}), \ldots, T(A_{1k}), \ldots, T(A_{mk})$ are pairwise vertex-disjoint, since otherwise the subgraphs $A_{11}, \ldots, A_{1k}, \ldots, A_{mk}$ would also be pairwise vertex-disjoint, yielding k disjoint H-expansions in G, a contradiction. Thus by (8.6) and (8.7) there exists $Y \subseteq V(T)$ with $|Y| < mk$ such that for some $j (1 \leq j \leq m)$, Y meets $T(A)$ for every $A \in \mathcal{A}_j$. Let $X = \bigcup(X_t : t \in Y)$. Then $X \cap V(A) \neq \emptyset$ for every H-expansion A in G, and

$$|X| \leq (mk-1)(w+1) = k'$$

as required.

We turn now to the converse of (8.8), to complete the proof of (8.2). A *surface* is a compact 2-manifold. If Σ is a connected orientable surface, we define its genus $g(\Sigma)$ as usual—that is, intuitively, the number of handles we must add to a sphere to construct Σ. We wish to extend this definition to disconnected orientable surfaces. Let Σ be an orientable surface, with components $\Sigma_1, \ldots, \Sigma_k$, numbered so that $g(\Sigma_1) \geq \cdots \geq g(\Sigma_k)$. We define $g(\Sigma)$ to be the sequence $g(\Sigma_1), \ldots, g(\Sigma_k)$.

We order surfaces by genus lexicographically; that is, if Σ, Σ' are orientable surfaces with

$$g(\Sigma) = g_1, \ldots, g_k$$
$$g(\Sigma') = h_1, \ldots, h_l,$$

we say $g(\Sigma) < g(\Sigma')$ if there exists j with $1 \leq j \leq \min(k, l)$, such that $g_j < h_j$ and $g_i = h_i$ $(1 \leq i < j)$. We assume the following three facts.

(8.9) *There is no infinite sequence $\Sigma_1, \Sigma_2, \ldots$ of orientable surfaces with $g(\Sigma_{i+1}) < g(\Sigma_i)$ $(i = 1, 2, \ldots)$.*

(8.10) *Let Σ be an orientable surface, and let $C \subseteq \Sigma$ be homeomorphic to a circle. Then one of the following holds:*

(i) *there is a closed disc $\Delta \subseteq \Sigma$ with $C = bd\ \Delta$;*

(ii) *there is an orientable surface Σ' with $g(\Sigma') < g(\Sigma)$, and disjoint closed discs $\Delta_1, \Delta_2 \subseteq \Sigma'$, such that $\Sigma' - (\Delta_1 \cup \Delta_2) = \Sigma - C$.*

[p. 113] (8.11) *If r is a connected open subset of a surface Σ, r is not a whole component of Σ homeomorphic to a sphere, and every subset of r homeomorphic to a circle bounds a closed disc in Σ, then there is an open disc $\Delta \subseteq \Sigma$ with $r \subseteq \Delta$.*[F7]

The paper assumes (8.11), as it says before (8.9). We prove the corrected statement. The set r lies in one component Σ_0 of Σ, since it is connected. A closed disc whose boundary lies in Σ_0 lies in Σ_0, since it is connected, and so the hypothesis holds with Σ_0 in place of Σ. Thus we may assume that Σ is connected. If Σ is a sphere, then $r \neq \Sigma$, and for any point $p \in \Sigma - r$ the set $\Sigma - \{p\}$ is an open disc including r. So we may assume that Σ is not a sphere. Let us call $Z \subseteq \Sigma$ *flat* if every closed curve in Z is null-homotopic in Σ. Then r is flat. For otherwise, by (11.2) of [12] (Graph Minors VII, J. Combin. Theory Ser. B 45 (1988), 212–254), some subset of r homeomorphic to a circle is not null-homotopic in Σ, but it bounds a closed disc in Σ by hypothesis, and so it is null-homotopic. By Zorn's lemma, r is included in a maximal set Y that is open, path-connected and flat. (The union of a chain of such sets containing r is again such a set, since a closed curve in it is compact and so lies in one member of the chain.) Now let $C \subseteq Y$ be homeomorphic to a circle. Since C is null-homotopic in Σ, it bounds a closed disc $D \subseteq \Sigma$, by a theorem of Epstein (D. B. A.

[F7] **Statement mistake.** As printed, (8.11) is false when r is a whole component of Σ that is a sphere. Then r is connected and open, and every circle in it bounds a closed disc, but r is not contained in an open disc. The corrected statement excludes exactly this case.

Epstein, Curves on 2-manifolds and isotopies, Acta Math. 115 (1966), 83–107, Theorem 1.7). The set $Y \cup D$ equals $Y \cup (D - C)$, and so it is open. It is path-connected, since Y and D are path-connected and both include C. It is flat by (11.3) of [12], applied with $\Delta = D$, since Y is flat and $Y \cap bd(D) = C$ is arc-connected. By the maximality of Y, we have $D \subseteq Y$. Thus every subset of Y homeomorphic to a circle bounds a closed disc included in Y, and by (4.2) of [12], Y is an open disc, since Σ is not a sphere. This proves (8.11).[F8]

Now let H be a graph. By (8.9), there is an orientable surface Σ in which H can be drawn, with $g(\Sigma)$ minimal.

(8.12) *For any drawing of H in Σ that meets every component of Σ homeomorphic to a sphere and any region r of this drawing, there is an open disc $\Delta \subseteq \Sigma$ with $r \subseteq \Delta$.*[F9]

Proof. If there is no such Δ, then by (8.11) there is a subset $C \subseteq r$ homeomorphic to a circle, which bounds no closed disc of Σ. But then by (8.10), there is an orientable surface Σ' with $g(\Sigma') < g(\Sigma)$ in which H can be drawn, a contradiction.

(8.13) *With H, Σ as in (8.12), any two drawings of H in Σ meet.*

Proof. Since H is non-planar, it has a non-planar component H^*. If there are two disjoint drawings of H in Σ, there is a drawing of H^* and a drawing of H which are disjoint. But this is impossible, by (8.12), since H^* is non-planar.

(8.14) *H does not have the Erdös–Pósa property.*

Proof. Let Σ be as in (8.12), and let $k' \geq 0$ be any integer. Take $2k' + 1$ drawings of H in Σ, chosen so that no point of Σ is in more than two of them, and only finitely many points of Σ are in more than one of them. If we view the crossings as representations of new 4-valent vertices, the union of these drawings is a drawing of a graph G. Now G has $2k' + 1$ subgraphs $H_1, \ldots, H_{2k'+1}$, each a subdivision of a copy of H, such that no vertex of G is in more than two of them. It follows that if $X \subseteq V(G)$ and $|X| \leq k'$

[F8] **Missing proof.** The paper assumes (8.11). The argument above is the one used in the formalization, which uses (4.2), (11.2) and (11.3) of Graph Minors VII and the theorem of Epstein.

[F9] **Statement mistake.** As printed, (8.12) can fail, because a surface with minimal $g(\Sigma)$ may have a spherical component that the drawing does not use. For example, a drawing of K_5 in the disjoint union of a torus and a sphere has the whole sphere as a region, and no open disc contains it. With the added hypothesis the proof goes through, using (8.10) and (8.11) as corrected. Where (8.12) is applied later, Σ should be chosen minimal and without components that the drawing does not use.

then $G \setminus X$ has an H-expansion; and yet G does not have two disjoint H-expansions, by (8.13), because G may be drawn in Σ. Thus H does not have the Erdös–Pósa property. We give the details, in three steps.

First, no graph drawn in Σ has two disjoint H-expansions. For if G' is drawn in Σ and has disjoint H-expansions A_1, A_2, then contracting edges of $A_1 \cup A_2$ shows that the graph consisting of two disjoint copies of H is isomorphic to a minor of G'. A minor of a graph drawn in Σ can also be drawn in Σ, and restricting such a drawing to the two copies gives two disjoint drawings of H in Σ, contrary to (8.13).

Second, we choose the drawings. Let $m = 2k' + 1$. Subdividing every edge of H once gives a graph J without loops, and H is isomorphic to a minor of J (contract one of the two halves of each edge). A drawing of H in Σ gives a drawing of J. Choose mutually disjoint closed discs R_w ($w \in V(J)$), with w in the interior of R_w, such that an edge of J meets R_w only if it is incident with w, and then in a simple curve from w to the boundary of R_w. For each edge e of J, with ends u and v, choose m mutually disjoint simple curves $M_1(e), \dots, M_m(e)$ running beside e, each with one end on the boundary of R_u, the other end on the boundary of R_v, and no other point in any of the discs, such that curves for distinct edges are disjoint. We call the ends of these curves *ports*, and so the ports on the boundary of each R_w are distinct. Now fix $w \in V(J)$, and view R_w as a round disc. For $i = 1, \dots, m$ in turn, choose a point x_i of the boundary of R_w, different from the ports and from $x_1, \dots, x_{i-1}$, and join x_i by straight chords to the ports of the curves $M_i(e)$ for the edges e incident with w. The point x_i and these chords form the ith *star* at w. A line meets the boundary circle in at most two points, and so a new chord lies on no line of an earlier chord. Only finitely many choices of x_i send a new chord through a point where two earlier chords meet, and we choose x_i to avoid them. Then two stars at w meet in only finitely many points, and no point is in three stars at w.

For $1 \le i \le m$, the ith stars at all the vertices of J, together with the curves $M_i(e)$ ($e \in E(J)$), form the ith drawing. No point of Σ is in more than two of these drawings, and only finitely many are in more than one, since the curves $M_i(e)$ are disjoint from each other and from the stars except at their ports. Let G be the graph drawn by their union, as above, with the points x_i, the ports, and the points in two drawings as vertices. For $1 \le i \le m$, each ith star is drawn by a connected subgraph of G, and each $M_i(e)$ is drawn by an edge of G joining the ith stars at the ends of e. Contracting the edges of the ith stars gives J, and so the union of the ith stars and the edges $M_i(e)$ is a J-expansion in G. Contracting also one of the two halves of each edge of H shows that it is an H-expansion H'_i in G,

all of whose vertices lie in ith stars. Hence no vertex of G is in more than two of $H'_1, \ldots, H'_m$, because it lies in some disc R_w, and no point of R_w is in three stars at w.

Third, let $X \subseteq V(G)$ with $|X| \leq k'$. Each vertex in X is in at most two of $H'_1, \ldots, H'_m$, and so X meets at most $2k' < m$ of them. Thus some H'_i is an H-expansion in $G \setminus X$. By the first step, G has no two disjoint H-expansions. So for $k = 2$, neither (i) nor (ii) in the definition of the Erdös–Pósa property holds for G, and this is true for every k'.[F10]

To sum up, (8.8) and (8.14) together yield (8.2), as required.

Another natural generalization of the Erdös–Pósa theorem would be to regard a circuit as a subdivision of a loop, rather than as a graph contractible to a loop. One could replace (i) and (ii) in the definition of the Erdös–Pósa property by

(i) G has k disjoint subgraphs, each a subdivision of a graph isomorphic to H

(ii) there exists $X \subseteq V(G)$ with $|X| \leq k'$ such that X meets every subgraph of G which is isomorphic to a subdivision of H.

[p. 114] We have been unable to decide if this property is again equivalent to planarity.

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[F10] **Missing proof.** The paper does not say how to choose the $2k' + 1$ drawings, and gives the last two steps briefly. The construction and the argument above are the ones used in the formalization.

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